

JPE Replication Report

2026-03-08

1 Paper Identification

ID	12345678
Author	Dr Prof James 'JayJay' Author
Title	Stand-in title
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

3 General

- ☐ Data Availability and Provenance Statements
 - ☐ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)

- ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☐ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
- ☒ List of tables and programs
- ☐ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. - As per the [JPE policy](#), “In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication.”

{REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

Input data sources used to build the analysis data.

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Correlates of War (CoW)	datas-tore/raw/cow	milita-sized_in-ter-state_dis-putes/MIDB_4.2.csv	Yes	No	Yes	Yes
Correlates of War (CoW)	datas-tore/raw/cow	milita-sized_in-ter-state_dis-putes/MID-LOCI_2.0.csv	Yes	No	Yes	Yes
Correlates of War (CoW)	datas-tore/raw/cow	wars/Inter-StateWarData_v4.0.dta	Yes	No	Yes	Yes

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Correlates of War (CoW)	datas-tore/raw/cow	wars/Extra-StateWarData_v4.0.dta	Yes	No	Yes	Yes
Correlates of War (CoW)	datas-tore/raw/cow	wars/Intra-StateWarData_v4.1.dta	Yes	No	Yes	Yes
Fraser Institute's Economic Freedom Index (EFI)	datas-tore/raw/economic_free-dom/	efi_data.dta	Yes	No	Yes	Yes
Global Financial Data (GFD/Fin-aeon)	datas-tore/raw/gfd/try_series/	coun-	No	Yes	Yes	Yes
Global Financial Data (GFD/Fin-aeon)	datas-tore/raw/gfd/ries_list.xlsx	gfd_data_series_list.xlsx	No	Yes	Yes	Yes
Global Financial Data (GFD/Fin-aeon)	datas-tore/raw/gfd/sions.csv	gfd_recessions.csv	No	Yes	Yes	Yes
Global Financial Data (GFD/Fin-aeon)	datas-tore/raw/gfd/	govt_rev_gdp_data	No	Yes	Yes	Yes
Global Financial Data (GFD/Fin-aeon)	datas-tore/raw/gfd/	supplements/extraTotalReturn-Clean.xlsx	No	Yes	Yes	Yes

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Global Financial Data (GFD/Finaeon)	datas-tore/raw/gfd/	supple-ments/price_earn-ings_clean.csv	No	Yes	Yes	Yes
International Crisis Behavior (ICB)	datas-tore/raw/icb/	icb2v12.csv	Yes	No	Yes	Yes
Jordà-Schularick-Taylor (JST) Macrohistory Database	datas-tore/raw/jst/	JST-datasetR5.dta	Yes	No	Yes	Yes
Maddison Historical Statistics	datas-tore/raw/ggdc/	mpd2020.dta	Yes	No	Yes	Yes
Penn World Table	datas-tore/raw/pwt/	pwt100.dta	Yes	No	Yes	Yes
Penn World Table	datas-tore/raw/pwt/	pwt100-capital-detail.dta	Yes	No	Yes	Yes
Penn World Table	datas-tore/raw/pwt/	pwt100-labor-detail.dta	Yes	No	Yes	Yes
Relative Political Capacity Dataset (RPC)	datas-tore/raw/arpc/	arpc_2020_country.dta	Yes	No	Yes	Yes
Standardized World Income Inequality Database (SWIID)	datas-tore/raw/swiid/	swiid9_0_summary.csv	Yes	No	Yes	Yes

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Varieties of Democracy (V-Dem)	datas-tore/raw/vdem6/	V-Dem-6Y+Others-v8.dta	Yes	No	Yes	Yes
Varieties of Democracy (V-Dem)	datas-tore/raw/vdem6/	V-Dem-6Y-Full+Others-v10.dta	Yes	No	Yes	Yes
Wharton Research Data Services (WRDS)	datas-tore/raw/wrds/	factset_an-nual_fiscal.dta	No	Yes	Yes	Yes
Wharton Research Data Services (WRDS)	datas-tore/raw/wrds/	ibes_global_annual.dta	No	Yes	Yes	Yes
World Religion Dataset (WRD)	datas-tore/raw/cow/religion/	WRP_national.csv	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datas-tore/raw/other_events/	rein-harvard/goff_defaults.dta	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datas-tore/raw/other_events/	rein-harvard/goff_financial_crisis.csv	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datas-tore/raw/other_events/	assassinations.dta	Yes	No	Yes	Yes

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Data from Published Papers and Other Events	datas-tore/raw/other_events/	HOG_death.csv	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datas-tore/raw/other_events/	country_events_to_iso3.dta	Yes	No	Yes	Yes
Country Code Crosswalks	datas-tore/raw/crosswalks/	country_name_to_iso3.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datas-tore/raw/crosswalks/	country_codes_to_iso3.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datas-tore/raw/crosswalks/	ibex_country_to_iso3_codes.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datas-tore/raw/crosswalks/	icb_country_code_to_iso3.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datas-tore/raw/crosswalks/	iso2_codes_to_iso3_codes.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datas-tore/raw/crosswalks/	region_map.dta	Yes	No	Yes	Yes
Country Code Crosswalks	datas-tore/raw/crosswalks/	valid_iso3_codes.dta	Yes	No	Yes	Yes
Foreign Direct Investment (FDI)	datas-tore/raw/fdi/flows_raw.csv	fdi_in-	Yes	No	Yes	Yes
Foreign Direct Investment (FDI)	datas-tore/raw/fdi/flows_raw.csv	fdi_out-	Yes	No	Yes	Yes

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Regime Change Data	datas-tore/raw/regime_change/	DDCG-national.dta	Yes	No	Yes	Yes
Regime Change Data	datas-tore/raw/regime_change/	ERT.csv	Yes	No	Yes	Yes
Regime Change Data	datas-tore/raw/regime_change/	historical_changes_for_democratization.csv	Yes	No	Yes	Yes
Regime Change Data	datas-tore/raw/regime_change/	vod_democratizations.csv	Yes	No	Yes	Yes
World Income Inequality Database (WIID)	datas-tore/raw/wiid/	wiidcountry.dta	Yes	No	Yes	Yes

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☐ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

{REQUIRED} Please review authors and affiliations on the deposit. In general, they are the same, and in the same order, as for the manuscript; however, authors can deviate from that order.

6.1 Stated Requirements

“The code was last run on a 10-core Intel-based laptop with WindowsOS version 11 Enterprise with 50GB of free space. Computation took 21 minutes”

- ☒ Software Requirements specified as follows:
 - Python ≥ 3.11 , < 3.13
 - Stata (code was last run with version 19)
- ☒ Computational Requirements specified as follows:
 - 2 GB - 25 GB required storage space
- ☒ Time Requirements specified as follows:
 - 10-60 minutes

7 Analyzing Data and Code Files

7.1 Code description

The author implements the replication using a SCons-based reproducible research pipeline, orchestrated through uv, which automatically runs the Python and Stata scripts needed to generate the paper’s tables and results.

There are 36 .do scripts which generate the analysis data.

There are 10 .do and 2 .py scripts which generate the main tables.

There are 7 .do and 1 .py scripts which generate the main figures.

There are 16 .do and 1 .py scripts which generate the appendix tables.

There are 11 .do scripts which generate the appendix figures.

Summarizing the Data Cleaning Code

source/derived/01_events/

Dataset Produced	Program
	01_create_valid_iso3_codes.do
	02_create_pre1900_dem.do
	03_create_regime_change.do
	04_create_coup_detat.do
	05_create_financial_crises.do
	06_create_sovereign_default.do
	07_create_hog_deaths.do
	08_create_international_crises.do
	09_create_recession.do
	10_create_interstate_wars.do

Dataset Produced	Program
	11_create_intrastate_wars.do
	12_create_extrastate_wars.do
	13_create_militarized_disputes.do
	14_combine_all_events.do

source/derived/02_macro_political/

Dataset Produced	Program
	01_create_inflation.do
	02_create_pwt_data.do
	03_create_gdp.do
	04_create_vdem_datasets.do
	05_create_portion_catholic.do
	06_create_gini.do
	07_create_tax_rates.do
	08_create_gov_rev.do
	09_create_economic_competition.do
	10_create_fdi.do

source/derived/03_assets/

Dataset Produced	Program
	01_factset_clean.do
	02_ibes_global_clean.do
	03_create_dividend_yields.do
	04_create_equity_returns.do
	05_create_fixed_income.do
	06_create_home_country_bond_rate.do
	07_create_excess_and_abnormal_returns.do
	08_create_cashflow_growth.do
	09_create_other_asset_market_variables.do
	10_create_return_news.do

source/derived/04_merging/

Dataset Produced	Program
	01_create_section_3_and_5_data.do

Dataset Produced	Program
	02_create_section_4_data.do

Summarizing code producing Tables and Figures:

Figure/Table #	Program	Filename	Reproduced?
Main Tables	source/analysis/main/tables/	source/analysis/main/tables/raw	
Table 1	01_table_1_change_in_log_dividend_yield.do	table_1_change_in_log_dividend_yields.tex	No
Table 2	02_table_2_cash-flow_growth.do	table_2_cash-flow_growth.tex	No
Table 3	03_table_3_democratization_vs_other_political_risk.do	table_3_democratization_vs_other_political_risk.tex	No
Table 4	04_table_4_revolution_risk.do	table_4_revolution_risk.tex	No
Table 5	05_table_5_regional_waves_instrument.do	table_5_regional_waves_instrument.tex	No
Table 6	06_table_6_balance_tests.do	table_6_balance_tests.tex	No
Table 7	07_table_7_did_results.do	table_7_did_results.tex	No
Table 8	08_table_8_explicit_redistribution.do	table_8_explicit_redistribution.tex	No
Table 9	09_table_9_implicit_redistribution.do	table_9_implicit_redistribution.tex	No
Table 10	10_table_10_high_vs_low_redistribution_risk.do	table_10_high_vs_low_redistribution_risk.tex	No
Table 11	11_table_11_model_calibration.py	table_11_model_calibration.tex	No
Table 12	12_table_12_model_results.py	table_12_model_results.tex	No

Main Figures	source/analysis/main/figures/	figures/	raw
Figure 1	01_figure_1_dividend_yield_event_study.do	figure_1_dividend_yield_event_study.pdf	No
Figure 2	02_figure_2_physical_human_capital.do	figure_2_physical_human_capital.pdf	No
Figure 3	03_figure_3_gdp_consumption_distributions.do	figure_3a_gdp_growth_distribution.pdf, figure_3b_consumption_growth_distribution.pdf	No
Figure 4	04_figure_4_regional_waves.do	figure_4_regional_waves.pdf	No
Figure 5	05_figure_5_anti_regime_event_study.do	figure_5a_event_study_anti_system_cso.pdf, figure_5b_event_study_democratic_protests.pdf	No
Figure 6	06_figure_6_returns_event_study.do	figure_6_event_study_returns.pdf	No
Figure 7	07_figure_7_dividend_yield_coefficients_over_time.do	figure_7_coefficients_over_time.pdf	No
Figure 8	08_figure_8_autocratization_figure.py	figure_8_autocratization_model.pdf	No
Appendix Tables	source/analysis/appendix/tables/	tables/	raw
Table A1	01_table_A1_summary_statistics.do	A1_summary_statistics.tex	No
Table B2	02_table_B2_adverse_dividend_growth.do	B2_adverse_dividend_growth.tex	No
Table B3	03_table_B3_risk_premium.do	B3_risk_premium.tex	No

Appendix Tables	source/analysis/appendix/tables/raw	source/analysis/appendix/tables/raw	source/analysis/appendix/tables/raw
Table B4	04_table_B4_macro_political_risk_measures.do	B4_macro_political_risk_measures.tex	No
Table B5	05_table_B5_adverse_probability.do	B5_adverse_probability.tex	No
Table B6	06_table_B6_democratize_risk_measures.do	B6_democratize_risk_measures.tex	No
Table C7	07_table_C7_probability_democratize_post_vatican_ii.do	C7_vatican_ii_democratize.tex	No
Table C8	08_table_C8_vatican_i.do	C8_did_first_vatican.tex	No
Table C9	09_table_C9_catholic_democracies.do	C9_did_democracies.tex	No
Table C10	10_table_C10_removing_outliers.do	C10_did_removing_outliers.tex, C10_did_removing_outliers_long.tex	No
Table C11	11_table_C11_outlier_ro-bust_weights.do	C11_did_ro-bust_weights.tex	No
Table C12	12_table_C12_global_capm.do	C12_did_global_capm.tex	No
Table C13	13_table_C13_no_rolling_beta.do	C13_did_no_rolling_beta.tex	No
Table C14	14_table_C14_home_country_bonds.do	C14_did_results_country_bonds.tex	No
Table C15	15_table_C15_capital_gains_control.do	C15_did_capital_gains.tex	No
Table D16	16_table_D16_inequality_price_decline.do	D16_inequality_price_decline.tex	No
Table G17	17_table_G17_ert_democracy.py	table_G17_democratizations_and_history.tex	No

Appendix Figures	source/analysis/appendix/figures/	source/analysis/appendix/figures/raw	
Figure B1	01_figure_B1_democracy_log_price.do	B1_democracy_log_price_dividend.pdf, B1_democracy_log_price_gdppc.pdf, B1_democracy_log_price_price.pdf	No
Figure C2	04_figure_C4_cso_activity_vs_mobilizations.do	C2_democracy_activity_mobil.pdf	No
Figure C3	07_figure_C7_country_pair_1946_1976.do	C3_country_pair_falsification_aut.pdf, C3_country_pair_falsification.pdf	No
Figure C4	06_figure_C6_window_end_date.do	C4_window_end_date_all.pdf, C4_window_end_date_aut.pdf	No
Figure C5	08_figure_C8_country_pair_1939_1983.do	C5_country_pair_PE_short_aut.pdf, C5_country_pair_PE_short.pdf, C5_country_pair_T_short_aut.pdf, C5_country_pair_T_short.pdf	No
Figure C6	08_figure_C8_country_pair_1939_1983.do	C6_country_pair_PE_long_aut.pdf, C6_country_pair_PE_long.pdf, C6_country_pair_T_long_aut.pdf, C6_country_pair_T_long.pdf	No
Figure C7	09_figure_C9_dividend_yield_event_study.do	C7_dividend_yield_event_study.pdf	No

Appendix Figures	source/analysis/appendix/figures	source/analysis/appendix/figures/raw	
Figure D8	10_figure_D10_explicit_redis-tribute_event_study.do	D8_explicit_redis-tribute_event_study_govt_rev_gdp.pdf, D8_explicit_redis-tribute_event_study_swild.pdf	No
Figure D9	11_figure_D11_democratization_end_price_response.do	D9_democratize_price_response_ld.pdf, D9_democratize_price_response_succ.pdf	No
Figure F11	12_figure_F13_sweden_case_study.do	F11_sweden_dividend_yield.pdf	No
Figure F12	13_figure_F14_france_case_study.do	F12_france_dividend_yield.pdf	No

Summarizing In-text Numbers

In-text numbers	Location in Manuscript	Program	Line Number	Reproduced?
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NOTE: In-text numbers that reference numbers in tables do not need to be listed.
Only in-text numbers that correspond to no table or figure need to be listed.

7.2 Computing Environment of the Replicator

- Mac Laptop M2 MacBook Air, MacOS 15.7.4, 16 GB of memory, (M2 Processor, 8 cores)
- Stata/SE 19.5.0
- VSCode 1.109.2 with Py3.13
- R 4.5.2 with RStudio 2026.01 with Quarto and JDK 21

8 Replication

Ran the following commands in bash to start the replication:

```
conda create -n Miller python=3.12
conda activate Miller
python -m pip install uv
python -m uv run scons
```

The below error was returned after the last command:

```
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$
m uv run scons
scons: Reading SConscript files ...
scons: done reading SConscript files.
scons: Building targets ...
build_python(["source/paper/numbers/section_3.tex"], ["source/analysis/numbers/02_section_3_1
scons: *** [source/paper/numbers/section_3.tex] ExecCallError : Python did not run successfu
Command tried: "python" source/analysis/numbers/02_section_3_numbers.py > source/paper/numb
Traceback (most recent call last):
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/analysis/n
    from source.config import ROOT_PATH
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/config.py"
    ROOT_PATH = Path(os.environ["ROOT_PATH"])
                        ~~~~~~^~~~~~
  File "<frozen os>", line 685, in __getitem__
KeyError: 'ROOT_PATH'

Traceback (most recent call last):
  File "/files/JPE-Miller-20240145/replication-package/replication_package/.venv/lib/python3
packages/SCons/Action.py", line 1441, in execute
    result = self.execfunction(target=target, source=rresources, env=env)
              ~~~~~~^~~~~~
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab
    builder.execute_system_call()
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab
    self.do_call()
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab
    self.raise_system_call_exception(traceback = traceback)
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab
    raise ExecCallError(message)
source.lib.JMSLab._exception_classes.ExecCallError: Python did not run successfully. Please
Command tried: "python" source/analysis/numbers/02_section_3_numbers.py > source/paper/numb
Traceback (most recent call last):
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/analysis/n
    from source.config import ROOT_PATH
```



```
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/config.py"
    ROOT_PATH = Path(os.environ["ROOT_PATH"])
    ~~~~~~^~~~~~
File "<frozen os>", line 685, in __getitem__
KeyError: 'ROOT_PATH'
```

scons: building terminated because of errors.

Which was fixed using:

```
export ROOT_PATH="$PWD"
python -m uv run scons
```

Which then returned the below error:

```
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$
m uv run scons
scons: Reading SConscript files ...
```

```
scons: *** missing SConscript file 'source/analysis/paper/SConscript'
File "/files/JPE-Miller-20240145/replication-package/replication_package/SConstruct", line 3:
```

paper/ is stored in source/, not under source/analysis/.

The replicator copies the the file and proceeds:

```
mkdir -p source/analysis/paper
cp -r source/paper/* source/analysis/paper/
```

The below error is returned. The build calls `pdflatex` but this is not listed in the computational requirements, nor included in the dependencies downloaded by the package.

Replicator stops replication here.

```
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$
m uv run scons
scons: Reading SConscript files ...
scons: done reading SConscript files.
scons: Building targets ...
build_python(["source/paper/miller_2025.pdf", "source/paper/online_appendix.pdf"], ["source/p
scons: *** [source/paper/miller_2025.pdf] ExecCallError : Python did not run successfully. P
```

Command tried: "python" source/paper/compile_paper.py > source/paper/sconscript.log

Traceback (most recent call last):

```
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/compile_paper.py", line 10, in
    main()
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/compile_paper.py", line 15, in
    compile_all()
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/compile_paper.py", line 20, in
    run_pdflatex(stem)
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/compile_paper.py", line 25, in
    run_command(
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/compile_paper.py", line 30, in
    result = subprocess.run(
               ~~~~~
File "/usr/lib/python3.12/subprocess.py", line 548, in run
    with Popen(*popenargs, **kwargs) as process:
         ~~~~~
File "/usr/lib/python3.12/subprocess.py", line 1026, in __init__
    self._execute_child(args, executable, preexec_fn, close_fds,
File "/usr/lib/python3.12/subprocess.py", line 1955, in _execute_child
    raise child_exception_type(errno_num, err_msg, err_filename)
FileNotFoundError: [Errno 2] No such file or directory: 'pdflatex'
```

Traceback (most recent call last):

```
File "/files/JPE-Miller-20240145/replication-package/replication_package/.venv/lib/python3.12/site-packages/SCons/Action.py", line 1441, in execute
    result = self.execfunction(target=target, source=rsources, env=env)
               ~~~~~
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/build.py", line 10, in
    builder.execute_system_call()
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/build.py", line 15, in
    self.do_call()
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/build.py", line 20, in
    self.raise_system_call_exception(traceback = traceback)
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/build.py", line 25, in
    raise ExecCallError(message)
```

source.lib.JMSLab._exception_classes.ExecCallError: Python did not run successfully. Please check the command and its arguments.

Command tried: "python" source/paper/compile_paper.py > source/paper/sconscript.log

Traceback (most recent call last):

```
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/compile_paper.py", line 10, in
    main()
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/compile_paper.py", line 15, in
    compile_all()
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/compile_paper.py", line 20, in
    run_pdflatex(stem)
```

```

    run_pdflatex(stem)
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/comp
    run_command(
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/paper/comp
    result = subprocess.run(
        ~~~~~~
File "/usr/lib/python3.12/subprocess.py", line 548, in run
    with Popen(*popenargs, **kwargs) as process:
        ~~~~~~
File "/usr/lib/python3.12/subprocess.py", line 1026, in __init__
    self._execute_child(args, executable, preexec_fn, close_fds,
File "/usr/lib/python3.12/subprocess.py", line 1955, in _execute_child
    raise child_exception_type(errno_num, err_msg, err_filename)
FileNotFoundError: [Errno 2] No such file or directory: 'pdflatex'

scons: building terminated because of errors.

```

9 Findings

Code did not start running as per the author's SCons-based pipeline. No findings can be reported as of yet.

9.1 Data Preparation Code

9.2 Tables and Figures

None of the tables and figures were reproduced.

9.3 In-Text Numbers

- ☒ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☐ partial reproduction (see above)
- ☒ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☒ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements